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# Exhaustive search for novel multicomponent alloys with brute force and machine learning

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We present an algorithm for the high-throughput computational discovery of intermetallic compounds in systems with a large number of components. It is particularly important for high entropy alloys (HEAs), where multiple principal elements can form numerous potential intermetallic compounds during the condensation process, making it challenging to predict the dominant phase. Our algorithm is based on a brute-force evaluation of candidate structures with a fixed underlying lattice (FCC or BCC) accelerated by machine-learning interatomic potentials. The algorithm takes a set of chemical elements and a crystal lattice type as inputs and produces structures on and near the convex hull of thermodynamically stable structures. The candidate structures are evaluated using the low-rank potential (LRP), trained to reproduce energies of structures equilibrated with density functional theory (DFT). Thanks to extreme computational effectiveness of the LRP, it is feasible to evaluate hundreds of thousands of structures per second, per CPU core. Thus, our algorithm screens a complete set of candidate structures for a given system without missing any configurations. We validated our method on systems with BCC (Nb-W, Nb-Mo-W, V-Nb-Mo-Ta-W) and FCC (Cu-Pt, Cu-Pd-Pt, Cu-Pd-Ag-Pt-Au) lattices and discovered 268 new alloys not reported in the AFLOW database<sup>1</sup>, which we used as a benchmark.

High-entropy alloys are a highly promising class of materials due to the stability of the solid solution phase in a wide range of compositions<sup>2</sup>. However, the large number of components not only reduces the free energy of the solid solution making it an entropically favorable phase, but also vastly increases the number of intermetallic compounds that compete with the solid solution for the dominant phase<sup>3</sup>. Therefore, being able to computationally discover all intermetallic compounds in a system with a large number of constituent elements is important for the computational study of high-entropy alloys.

The rapid development of computational technologies in materials science over the past two decades enabled the discovery and design of new materials on a computer. With this approach, simulation and even discovery of materials are typically based on first-principles methods, such as Density Functional Theory (DFT)<sup>4</sup>, which offer sufficient reliability for making predictions. Experimental synthesis is then performed only at the final stage and only for the most promising candidates<sup>5–7</sup>—this significantly accelerates the search and design, thus reducing its cost.

Computer-driven materials discovery involves predicting crystal structures with DFT-calculated formation energy on the convex hull or in its close proximity. In the general case, this problem can be formulated as finding several global minima on the potential energy surface in the

multidimensional space of atomic positions, their types, and lattice vectors. The key challenge here is that the number of local minima, corresponding to high-energy structures, grows exponentially with the number of atoms and atomic types<sup>8</sup>. For a system with specified chemical elements, during the condensation process, various potential intermetallic compounds can be formed, making it challenging to predict which specific phase will dominate.

State-of-the-art methods addressing this problem employ comprehensive approaches based on structure generation algorithms (for example, PyXtal<sup>9</sup>) and global optimization techniques (e.g. evolutionary algorithms<sup>9</sup>, particle swarm optimization<sup>10</sup>, metadynamics<sup>11</sup>). However, these methods are unable to perform an exhaustive search, and generally do not guarantee finding the material, even if it does exist. Moreover, their efficiency dramatically decreases in practice if the number of chemical elements in the system is larger than two.

Recently, a substantial acceleration in crystal structure prediction has been achieved by using machine-learning interatomic potentials (MLIPs) such as Neural-network potentials<sup>12,13</sup>, Moment Tensor Potential (MTP)<sup>14</sup>, Gaussian Approximation Potential (GAP)<sup>15</sup>, etc. MLIPs are typically trained on DFT-calculated structures and are able to reproduce the behavior of DFT with the computational efficiency being close to that of empirical potentials. These ideas enabled the solution of many problems of atomistic modeling

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such as predicting the stability of new materials<sup>12</sup>, understanding complex reaction pathways<sup>13</sup>, and accurately simulating large molecular systems<sup>16,17</sup>. In particular, using MLIPs instead of DFT for crystal structure equilibration accelerates the search and predictions by several orders of magnitude<sup>18,19</sup>. However, even with MLIPs exhaustive search remains computationally unfeasible in the general case.

The complexity of the search can be further reduced if all the potential structures have the same lattice type known a priori (for example, in alloys which in most cases have either BCC, FCC, or HCP lattice). In this case, there is no need to vary positions of atoms in space, and the main efforts are applied to find the right order of atomic types in the sites of the underlying crystal lattice. One of the important works utilizing a lattice-based concept for discovery of new alloys and convex hull construction was proposed by G.W. Hart et al.<sup>5,20,21</sup> which became a basic method for gathering of AFLOW database<sup>1</sup>. Its principle consisted of generating (enumerating) a full set of symmetrically unique structures, constructed on a fixed crystal lattice. However, the proposed methodology involved computationally expensive equilibration (relaxation) originally done with DFT and later accelerated with MLIPs<sup>19</sup>. An alternative to structure enumeration was proposed by V. Sotnikov et al.<sup>22</sup>, in this algorithm the low-energy structures are generated dynamically. The algorithm featured a novel crystal structure “growing” approach in an atom-by-atom fashion, selecting the energetically favorable prototype on each step employing machine-learning potentials. However, this algorithm does not explore the phase space in an exhaustive manner and is unable to provide a complete convex hull.

This work is the result of our efforts aimed at the invention of a computationally efficient method to perform an exhaustive search for alloys and construction of convex hulls for a given metallic system. The method we propose is based on four key ideas: (1) restricting the search for a certain type of the crystal lattice (e.g. BCC, FCC, and HCP) that can be known (or expected) a priori; (2) using computationally efficient machine learning models of interatomic interaction; (3) performing an exhaustive search using brute-force algorithm; (4) updating the interatomic interaction model during the search in order to improve the accuracy and avoid extrapolation. In our approach, we do not equilibrate all the candidate structures appearing during the search. Instead, we use an interatomic interaction model designed and fitted to predict the energy of the equilibrated structures based on unequilibrated atomic positions. This feature provides enough computational efficiency to enable exhaustive search for alloys with two to five principal components and a reasonable number of atoms in the structure. The proposed methodology was tested by applying it to the discovery of new alloys and construction of convex hulls in the BCC-system V-Nb-Mo-Ta-W, the FCC-system Cu-Pd-Ag-Pt-Au, and their subsystems. The performance of the algorithm is evaluated by comparing our results to those obtained with high-throughput DFT calculations from the AFLOW database.

## Results

With the developed method, we performed an exhaustive search of two-, three-, and five-component alloys and constructed convex hulls for (1) BCC system V-Nb-Mo-Ta-W (refractory metals) and (2) FCC system Cu-Pd-Ag-Pt-Au (based on noble metals). For both systems, density functional theory (DFT), implemented in the Vienna Ab-initio Simulation Package (VASP)<sup>23</sup>, was used as a first principles model, specifically, the generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional<sup>4</sup>.

All DFT calculations (for training set creation, accuracy control, and validation of the search results) were performed with the following parameters: plane-wave cutoff energies of 500 eV, the smallest allowed spacing between k-points was not less than  $0.15 \text{ \AA}^{-1}$ , an energy tolerance of  $10^{-4}$  as a convergence criterion for the electronic iterations, and  $10^{-3}$  as a stopping criterion for structure equilibration.

### Results for the Nb-W BCC alloys

We started to explore practical applications of our method by testing the convex hull construction for the binary Nb-W system. Initially, the

functional form of our low-rank potential involved the dependence only on the 8 nearest neighbors, with the maximum rank of the TT-representation equal to  $r = 4$ .

The primary training set consisted of 100 atomic configurations with a BCC lattice, where two of them were monatomic pure metals and the rest were randomly generated.

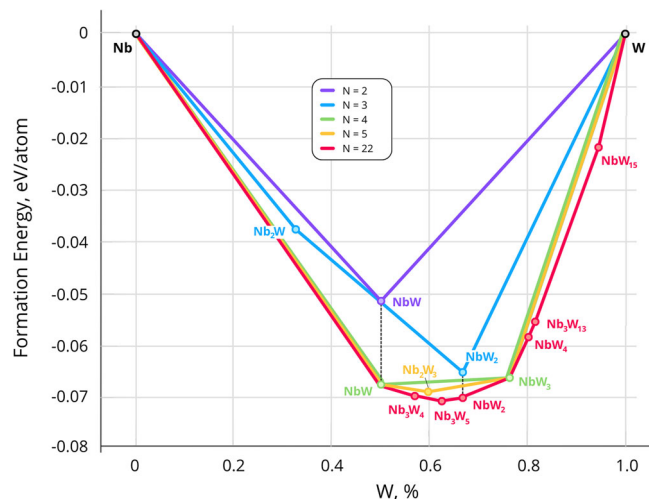
Furthermore, configurations with each number of atoms between 2 and 16 were guaranteed to be included in the training set. Here and in what follows, we preserved in the training set only topologically non-equivalent structures by comparing each new configuration with ones already in the set using the equivalence conditions, which are based on a simple comparison of (1) compositions and (2) per-atom energies of the structures. All generated structures for the training set were relaxed with DFT, which allowed the fitted LRP to implicitly reproduce the energy of the equilibrated structures without the need for relaxation of found configurations.

We estimated the accuracy of the first LRP version on a validation set of 200 random configurations equilibrated with DFT, resulting in the following validation errors: the mean absolute error (MAE) was 1.3 meV/atom, the root-mean-squared error (RMSE) was 1.4 meV/atom, and the maximum absolute error (MaxAE) was 7.0 meV/atom. In addition, in order to find dataset samples on which our model performed poorly, we created an ensemble of 10 LRPs, each of which differs from the original LRP by removing 10 random configurations from the initial training set. Using the obtained ensemble, we evaluated energies for all possible  $2^{16}$  configurations with 16 atoms in the supercell (not taking symmetries into account). The first 20 configurations that had the highest variation were relaxed by DFT and then added to the validation set. Finally, after re-fit of the obtained potential, 10 configurations that yielded the highest errors on the validation set were added to the training set. This idea allowed us to reduce MaxAE from 7.0 to 4.9 meV/atom, keeping MAE and RMSE at approximately the same level as before.

After the potential had been fitted, we began the search for structures using the brute-force method, starting from two atoms in the supercell. The layer value above the convex hull in which we considered structures was chosen to be approximately equal to the MaxAE or three times the RMSE. Shortly after, we found a NbW structure, which allowed us to construct an initial approximation of the convex hull. During the search process, we continued increasing  $N$ , gradually updating the convex hull after each iteration, having 103 non-equivalent structures in the  $\delta = 5$  meV layer above the convex hull when  $N$  reached 10. The search for up to 10 atoms in a supercell took about 40 seconds on a single CPU, while for up to 22 atoms in a supercell about 5 days was needed. The convex hull was updated 11 times during the search, and the last time this happened was when  $N = 9$ , and after that, the convex hull remained unchanged. Finally, when the number of atoms in supercell reached 22, the number of unique structures within the convex hull layer was 8558.

As noted above, during the search, our method can still find a few equivalent structures. To exclude identical configurations, we initially employed a comprehensive two-stage approach based on quick calculation of fingerprints (i.e. comparison of compositions and formation energies obtained by LRP), while for structures with the same fingerprints an additional, more careful check was carried out using hash calculations<sup>5</sup>. However, later we found that screening based on fingerprint calculation is sufficiently reliable for our purposes with a negligible chance to exclude the unique structure from search results. Therefore, since it had a significant performance advantage over the comprehensive method, we left only that first stage of screening, since it seemed to be optimal when comparing billions of structures.

After completing the search, we started to validate the results by conducting relaxation of the found structures with DFT. It was impossible to recalculate all the 8558 found structures from  $\delta = 5$  meV layer above the convex hull (and in the case of structures with more than two components this number is several orders of magnitude higher), so only the structures on convex hull, 150 closest to it, and 150 random structures within a 5 meV/atom gap were considered. As a result of the recalculation, we (1)



**Fig. 1 | Evolution of the Nb-W convex hull with increasing supercell size.** As the number of atoms in a supercell increases ( $N = 2, 3, \dots$ ), more new stable alloys are discovered, and the convex hull deepens.

constructed the DFT-validated convex hull and (2) estimated the error of our potential on low-energy structures. Moreover, the RMSE turned out to be 3 meV/atom, while the maximal absolute error (MaxAE) was about 12 meV/atom. Although this error is considered insignificant in many applications of computational materials science, it was essential to further improve the accuracy of LRP in order to accelerate screening by minimizing the number of structures falsely identified as stable during the search process.

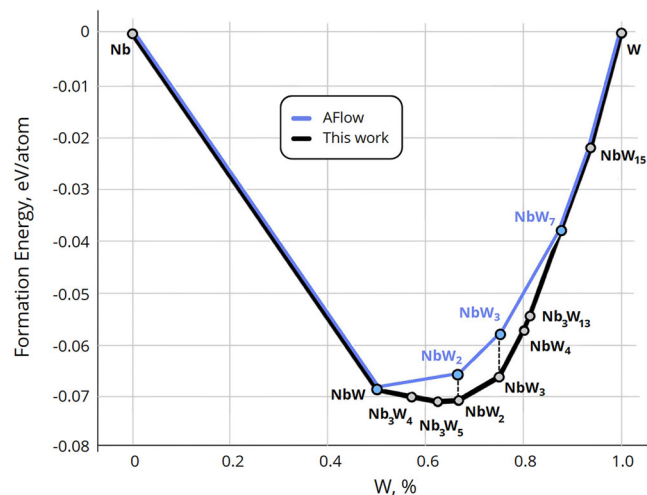
To further improve the LRP accuracy, we compared two energy values derived from LRP and DFT calculations for each structure selected after the search, thus, adding the top 40 structures with the highest discrepancy to the training set. Next, we increased the maximal rank of the TT-representation from  $r = 4$  to  $r = 6$  (further increase was not sufficiently efficient) and expanded the cutoff sphere of LRP by taking into account the second 6 nearest neighbors to the functional form. Therefore, the updated LRP depended on types of 15 neighbors (including the central atom type), while the initial LRP involved only 9 atomic types.

After combining these efforts, RMSE and MaxAE were reduced to 0.3 and 1.6 meV/atom, respectively, which allowed us to reduce the gap of interest for the search from 5 to 2 meV/atom, thus enhancing the calculation speed without increasing the chances to miss relevant structures. Then, we repeated all previous steps of the search with the updated LRP, where we evaluated all structures with  $N \leq 22$ , however starting from  $N = 16$  the convex hull remained unchanged (example of the convex hull updates for  $N = \{3, 4, 22\}$  atoms in the supercell is shown in Fig. 1). The number of found structures was 4021, which is still high, so we equilibrated with DFT only part of them (namely, all structures on the convex hull with 200 random structures from the 2 meV/atom layer above it).

Our method does not involve any a priori information by design, but in order to validate our results, we recalculated energies of alloy structures from the AFlow database. We found out that even in the case of two-component Nb-W system our DFT-validated convex hull contains 7 new structures, which are missing in AFlow (Fig. 2).

### Results for the ternary Nb-Mo-W BCC system

After completing the search in the binary system, we started to seek new alloys in the ternary Nb-Mo-W system using a workflow similar to that described earlier. This time, our LRP had a rank of 7 and included the 15 nearest neighbors within the cutoff radius, whereas the initial training set contained 120 configurations with each number of atoms in the supercell up to  $N = 16$ . We repeated the search three times, each time reaching  $N = 14$  atoms in the supercell. At the end of each iteration, after obtaining the



**Fig. 2 | Comparison of convex hulls for the Nb-W binary system.** The blue curve represents the convex hull we recalculated for the structures present in the AFlow database, used as a benchmark, while the black curve shows the convex hull for alloys discovered in this paper.

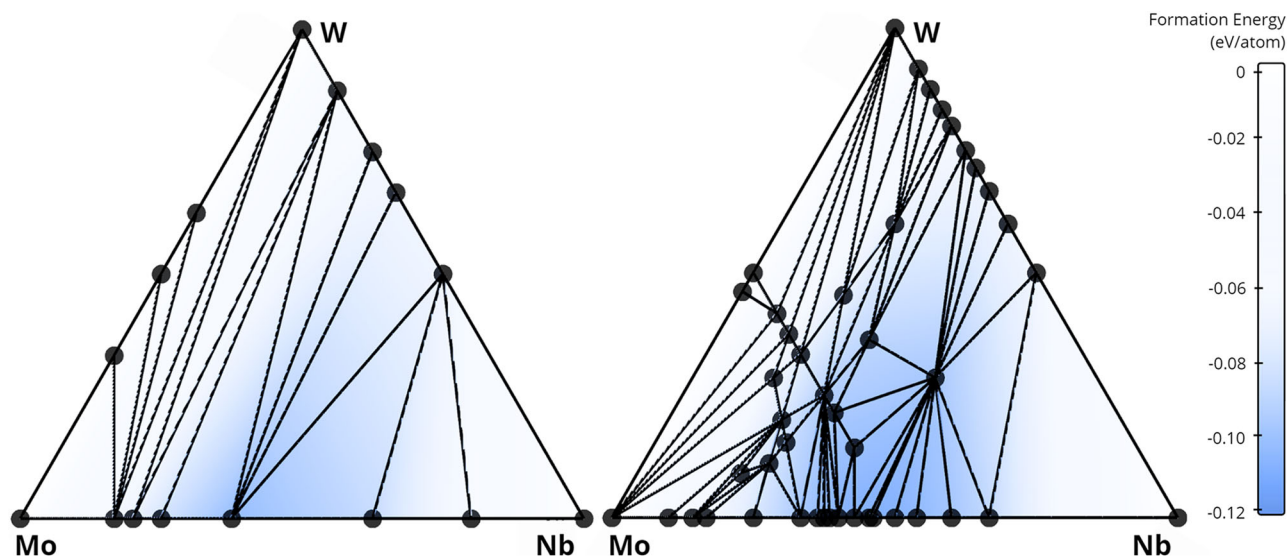
LRP-generated convex hull, we recalculated, using DFT, all structures that lay exactly on the convex hull and 100 random structures from a fixed layer above it. As mentioned previously, to enhance the accuracy of LRP, DFT-recalculated structures, in which energy calculations LRP demonstrated large errors, were added to the training set. The final version of LRP for Nb-Mo-W system had the RMSE of 0.3 and the MaxAE of 1.4 meV/atom, respectively. As a result, our DFT-calculated convex hull contained 12 new ternary alloys compared to the AFlow database, as shown in Fig. 3. Hereinafter, when constructing the final convex hull for systems with more than two components, we used all previously found structures from smaller subsystems. In this way, in the ternary Nb-Mo-W system, binary Nb-W structures with up to 22 atoms per supercell were included.

### Results for the five-component V-Nb-Mo-Ta-W BCC system

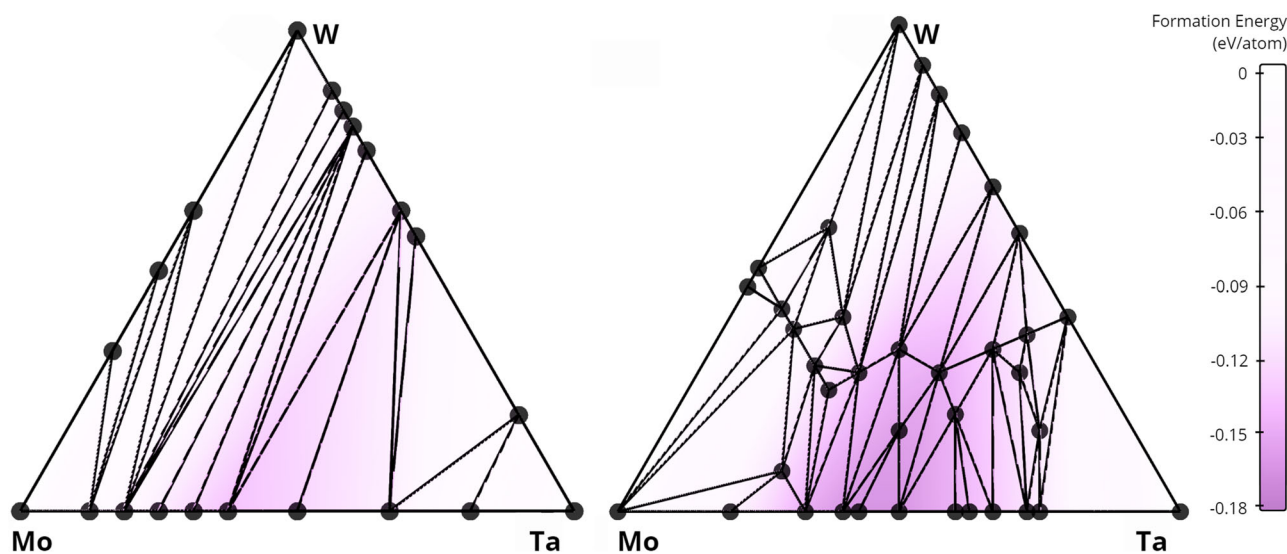
Next, we switched to finding new alloys in the five-component system V-Nb-Mo-Ta-W, where the initial training set consisted of 220 configurations. The search workflow, involving DFT-validation of the results and improving the LRP accuracy, was similar to that of binary and ternary systems and involved a total of 4 iterations. The final five-component LRP had the following errors after the last iteration: RMSE of 0.9 meV/atom and MaxAE of 4.0 meV/atom.

In total, we found 69 new binary and 45 new ternary alloy structures on the convex hull (DFT-proved) compared to the AFlow database (unfortunately, AFlow database is limited by three-component systems). For instance, we found 16 three-component alloys in the Mo-Ta-W system, whereas the AFlow database contains only two-component alloys for this system (see Fig. 4). Even though we did not find any five-component alloys on the convex hull, the best candidate lies within 5 meV/atom proximity to it, which is  $\text{VNbMo}_5\text{TaW}_3$ . In spite of that, our DFT-validated convex hull contains as many as 39 quaternary alloys. The full list of found structures on the convex hull with corresponding files can be found in supplementary data files (Section 5).

The maximum number of atoms in the supercell we reached during the search in this five-component system was 12. Since resource consumption increases exponentially with the number of atoms in a supercell, our brute-force method approaches its limits for practical use. For instance, in the case of the two-component BCC system, the search algorithm could reach 15 atoms in the supercell within 1 hour, whereas in the case of five components, this number decreases to 7 atoms (Fig. 5).



**Fig. 3 | Comparison of convex hulls for the Nb-Mo-W ternary system.** The left plot shows the recalculated convex hull for alloys from the AFLOW database, used as a benchmark, while the right plot shows the convex hull for alloys discovered in this paper.



**Fig. 4 | Comparison of convex hulls for the Mo-Ta-W ternary system.** The left plot shows the recalculated convex hull for alloys from the AFLOW database, used as a benchmark, while the right plot shows the convex hull for alloys discovered in this paper.

### Search for alloys in the Cu-Pd-Ag-Pt-Au system

After searching in BCC systems, we decided to test our method with FCC alloys. Unlike two- and three-component FCC systems, for a five-component system only the first coordination sphere, which contains 12 atoms, was considered, hence in that case our LRP depended on 13 atomic types (12 nearest neighbors plus the type of the central atom). Since this value is close to the number of atoms in the second coordination sphere of BCC systems, the time expenditure for the search in five-component FCC and BCC cases is similar (Fig. 5).

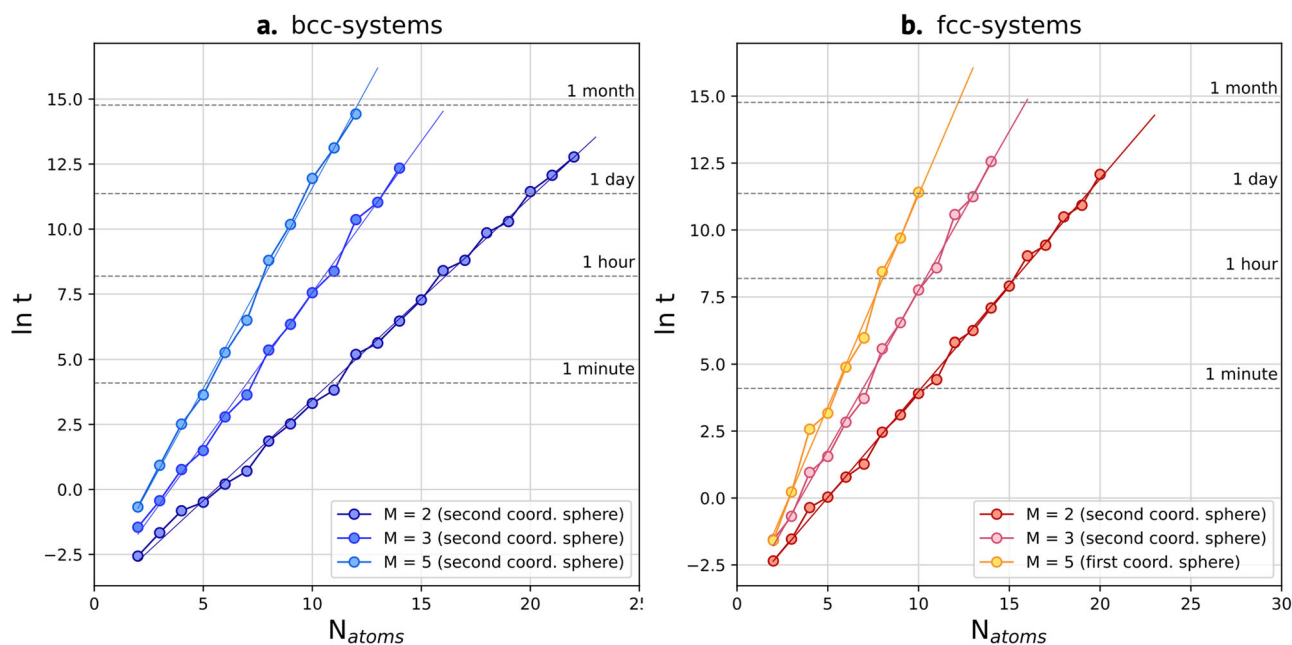
We performed the search successively, starting with the binary Cu-Pt system, then moving to the Cu-Pd-Pt ternary one, and finally finishing with the five-component Cu-Pd-Ag-Pt-Au system, while the workflow was again similar to the case of BCC systems. Namely, the initial numbers of configurations in the training sets were 30, 180, and 330 configurations for two-, three-, and five-component systems, respectively. After each search iteration, the results were partially recalculated using DFT and used to update LRP, improving its accuracy. The errors for the final LRPs of all FCC systems are

shown in Table 1, and they are slightly higher than those observed in BCC cases.

As a result, for the Cu-Pt, Cu-Pd-Pt, and Cu-Pd-Ag-Pt-Au systems, we found 76, 1434, and 14386 alloys respectively in the 3, 3, and 9 meV/atom layers above the corresponding convex hulls, while the difference in the gap values was related to the accuracy of each LRP. Moreover, we found 23 two-component, 74 three-component, and 18 four-component alloys on the convex hull, which are missing from the AFLOW database. A comparison of AFLOW and our convex hulls for the Cu-Pd-Pt and Pd-Ag-Au systems is shown in Figs. 6 and 7, respectively. Although we did not find any five-component alloys on the convex hull, performing the search for a longer time or further improving the LRP accuracy might enhance the results. The constructed convex hulls and corresponding structures are listed in supplementary data files (Section 5).

### Discussion

In this work, we introduced a method for the computational discovery of alloys at zero temperature. The method is based on a machine-



**Fig. 5 | Dependence of the algorithm's runtime on supercell size.** The horizontal axis indicates the maximum number of atoms in the supercell achieved during the search, and the vertical axis shows the natural logarithm of runtime for a body-

centered cubic and **b** face-centered cubic systems. Different lines represent variations in the number of atom types or cutoff parameters of low-rank potential.

**Table 1 | Training and validation errors of low-rank potentials in meV/atom**

|       |       | BCC systems |       |          | FCC systems |        |            |
|-------|-------|-------------|-------|----------|-------------|--------|------------|
|       |       | NbW         | NbMoW | VNbMoTaW | CuPt        | CuPdPt | CuPdAgPtAu |
| Train | RMSE  | 0.7         | 0.3   | 1.2      | 0.9         | 0.9    | 2.4        |
|       | MaxAE | 3.4         | 1.6   | 7.0      | 4.4         | 5.0    | 14.3       |
| Valid | RMSE  | 0.3         | 0.3   | 0.9      | 0.6         | 0.5    | 1.1        |
|       | MaxAE | 1.6         | 1.4   | 4.0      | 2.9         | 2.3    | 6.7        |

learning on-lattice model for evaluating formation energies and a brute-force algorithm for generating structures by varying atom types at the lattice sites. Our approach is limited to a fixed type of crystal lattice, which allows us to achieve superior computational efficiency. Its distinctive feature is the exhaustive search (at least up to some limited number of atoms in the supercell) that guarantees no alloys are missed, thus enabling the construction of the full convex hull for the system under consideration.

We have tested the developed method by performing a search for two-, three-, and five-component alloys in BCC (V-Nb-Mo-Ta-W) and FCC (Cu-Pd-Ag-Pt-Au) systems. By verifying the results with DFT, we found 268 different alloys whose formation energies are below the corresponding convex hulls from the AFlow database. It should be noted that to perform a correct comparison with AFlow, we recalculated all the structures from the database and the results of our search with the same settings and accuracy parameters of DFT. We also note that the AFlow database contains only structures with up to three components, while the convex hulls obtained in this paper were constructed for four- and five-component alloys. However, even in two- and three-component systems, we found alloys with formation energies more than 50 meV/atom lower than those on the convex hulls from AFlow.

Finally, the updated convex hull for the V-Nb-Mo-Ta-W system contains 69 two-component, 45 three-component, and 39 four-component new alloys not present in the AFlow database. Meanwhile, the results for the Cu-Pd-Ag-Pt-Au system include 23 two-component, 74 three-component, and 18 four-component new alloys.

The maximum number of atoms in a supercell that can be achieved during the search process strongly depends on the number of atomic types in the system. For two-component systems, we reached 22 atoms in a supercell, whereas for five-component systems, we reached 12 atoms in a supercell. Despite this difference, the time spent on the search in both cases was approximately the same. Therefore, since a five-component system contains 10 smaller subsystems (i.e., with fewer chemical elements), it is highly reasonable to perform the search separately in each of them, thus reaching a higher number of atoms in the supercell.

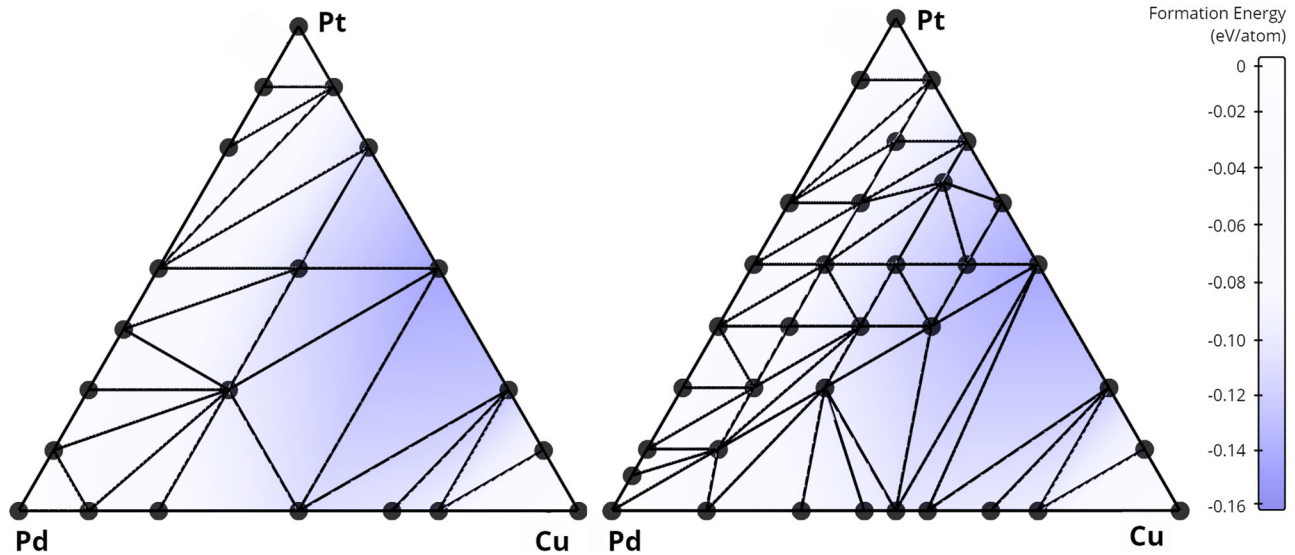
Perhaps the main difficulty associated with the practical application of our approach is the high number of candidate alloys in the layer above the convex hull due to the LRP uncertainty. However, this issue can be addressed by using more computational resources.

We believe that the systematic use of our method to search for new structures among various combinations of metal elements in the periodic table will stimulate the identification of multiple new alloys with unique features.

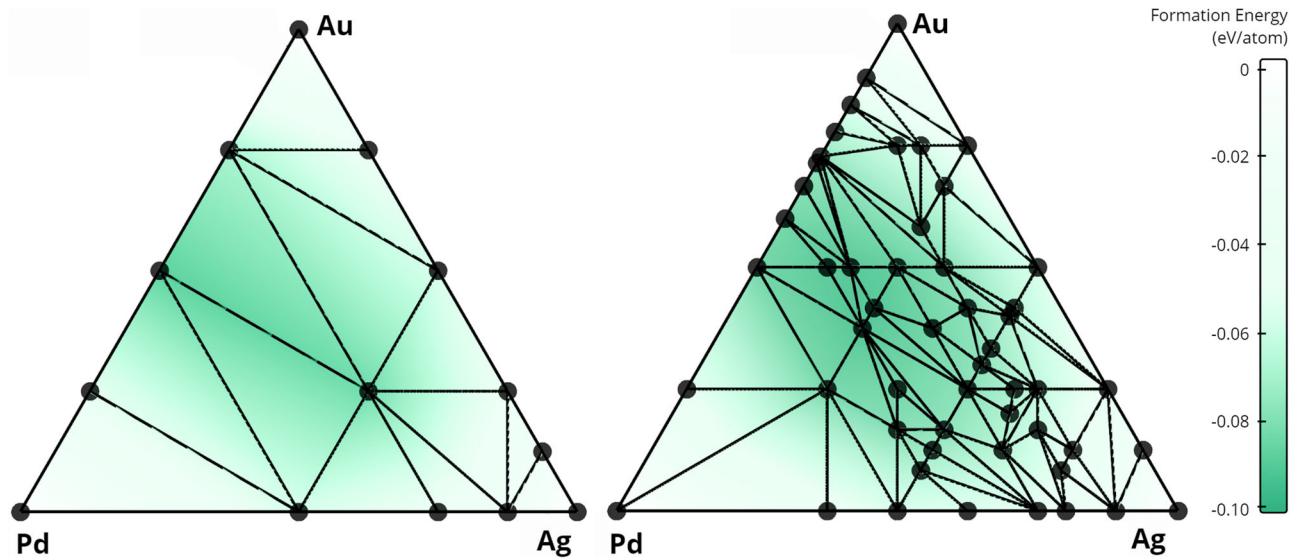
## Methods

### Brute-force method

We assume that, in the general case, the crystal lattice is represented by three linearly independent lattice vectors,  $\mathbf{l}_1$ ,  $\mathbf{l}_2$ ,  $\mathbf{l}_3$ , defining periodicity (lattice supercell), and coordinate vectors defining the positions of the sites  $\mathbf{c}_1, \dots, \mathbf{c}_k$  (understood as sublattices in fractional coordinates). In particular, matrices for the BCC lattice  $\mathbf{L}_{\text{bcc}}$  and the FCC lattice  $\mathbf{L}_{\text{fcc}}$  can be represented as



**Fig. 6 | Comparison of convex hulls for the Cu-Pd-Pt ternary system.** The left plot shows the recalculated convex hull for alloys from the AFLOW database, used as a benchmark, while the right plot shows the convex hull for alloys discovered in this paper.



**Fig. 7 | Comparison of convex hulls for the Pd-Ag-Au ternary system.** The left plot shows the recalculated convex hull for alloys from the AFLOW database, used as a benchmark, while the right plot shows the convex hull for alloys discovered in this paper.

follows:

$$\mathbf{L}_{\text{bcc}} = \{\mathbf{l}_1 = (-a/2, a/2, a/2), \mathbf{l}_2 = (a/2, -a/2, a/2), \mathbf{l}_3 = (a/2, a/2, -a/2), \mathbf{c}_1 = (0, 0, 0)\}, \quad (1)$$

$$\mathbf{L}_{\text{fcc}} = \{\mathbf{l}_1 = (0, a/2, a/2), \mathbf{l}_2 = (a/2, 0, a/2), \mathbf{l}_3 = (a/2, a/2, 0), \mathbf{c}_1 = (0, 0, 0)\}, \quad (2)$$

where  $a$  is a lattice parameter. An example of a lattice that requires at least two zero-coordinate vectors to define its matrix is the HCP lattice:

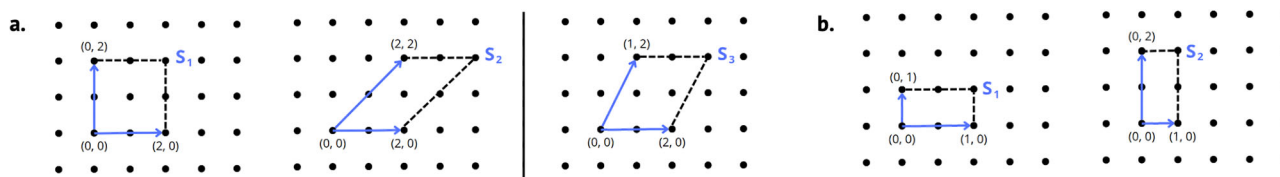
$$\mathbf{L}_{\text{hcp}} = \{\mathbf{l}_1 = (a, 0, 0), \mathbf{l}_2 = (-a/2, \sqrt{3}/2 a, 0), \mathbf{l}_3 = (a/2, a/2, 0), \mathbf{c}_1 = (0, 0, 0), \mathbf{c}_2 = (a/2, a/2, a/2)\}. \quad (3)$$

By the alloy supercell, we understand three linearly independent vectors  $\mathbf{s}_1$ ,  $\mathbf{s}_2$ , and  $\mathbf{s}_3$ , each of which is a linear combination of the lattice vectors  $\mathbf{l}_1$ ,  $\mathbf{l}_2$ , and  $\mathbf{l}_3$  with integer coefficients, which can be written in matrix form as:

$$\mathbf{S} = \begin{pmatrix} \mathbf{s}_1 \\ \mathbf{s}_2 \\ \mathbf{s}_3 \end{pmatrix} = \begin{pmatrix} z_{11} & z_{12} & z_{13} \\ z_{21} & z_{22} & z_{23} \\ z_{31} & z_{32} & z_{33} \end{pmatrix} \cdot \begin{pmatrix} \mathbf{l}_1 \\ \mathbf{l}_2 \\ \mathbf{l}_3 \end{pmatrix} = \mathbf{Z} \cdot \mathbf{L}, \quad (4)$$

where  $\mathbf{Z} \in \mathbb{Z}^{3 \times 3}$ . The number of atoms in the structure is defined by the supercell volume and is equal to  $|\det(\mathbf{Z})|$ .

Suppose we want to consider all possible structures with a specified crystal lattice for  $M$  given chemical elements up to a fixed number of atoms in the supercell  $N_{\text{max}}$ . For example, we aim to identify the most stable configurations from Nb and W ( $M = 2$ ) among structures with a



**Fig. 8 | Examples of equivalent supercells.** **a** The supercells in the left  $S_1$  and in the middle  $S_2$  are equivalent, and neither of them is equivalent to the supercell  $S_3$  in the right:  $|\det(S_2 S_1^{-1})| = 1$  and  $|\det(S_3 S_1^{-1})| = 0.5$ . **b** Equivalent supercells due to symmetries of the lattice itself.

**Table 2 | Number of non-equivalent supercells for different number of atoms in it for body-centered cubic and face-centered cubic lattices**

| $N_{\text{atoms}}$        | 1  | 2   | 3   | 4   | 5  | 6   | 7  | 8   | 9  | 10  | 11 | 12  |
|---------------------------|----|-----|-----|-----|----|-----|----|-----|----|-----|----|-----|
| Non-equivalent supercells | 1  | 2   | 3   | 7   | 5  | 10  | 7  | 20  | 14 | 18  | 11 | 41  |
| $N_{\text{atoms}}$        | 13 | 14  | 15  | 16  | 17 | 18  | 19 | 20  | 21 | 22  | 23 | 24  |
| Non-equivalent supercells | 15 | 28  | 31  | 58  | 21 | 60  | 25 | 77  | 49 | 54  | 33 | 144 |
| $N_{\text{atoms}}$        | 25 | 26  | 27  | 28  | 29 | 30  | 31 | 32  | 33 | 34  | 35 | 36  |
| Non-equivalent supercells | 50 | 72  | 75  | 123 | 49 | 158 | 55 | 177 | 97 | 112 | 99 | 268 |
| $N_{\text{atoms}}$        | 37 | 38  | 39  | 40  | –  | –   | –  | –   | –  | –   | –  | –   |
| Non-equivalent supercells | 75 | 136 | 129 | 286 | –  | –   | –  | –   | –  | –   | –  | –   |

BCC lattice with up to 20 atoms in a supercell. For every  $N \in [1, N_{\text{max}}]$  atoms in a supercell, there are  $N^M$  possible combinations of atomic types in the supercell without structure symmetries taken into account. In our research, we implement a brute force algorithm to generate all possible structures, evaluating the formation energy of each to find the most stable ones. Notably, equivalent structures are not entirely excluded from consideration, unlike in<sup>5</sup>, since in our case identification of equivalent structures is much more expensive than their energy evaluation.

To construct a convex hull and find all potentially stable alloy structures we need not only to vary the atomic types in the sites of a supercell, but also to find all non-equivalent supercells with different number of atoms (i.e., different volumes). In our method it is done via three nested loops, as illustrated in Algorithm 1.

#### Algorithm 1. Brute-force method

```

1: for do
2:   Supercells = GenerateNonequivalentSupercells(N)
3:   for each supercell of Supercells do
4:     for each TypeCombination of GetAllPossibleTypeCombinations
      (N, supercell) do
5:       EvaluateEnergy(N, supercell, TypeCombination)
6:     end for
7:   end for
8: end for

```

The algorithm is implemented using a mixture of C++ and Python, with Cython as the interface between the two parts of the code. The most computationally intensive part—the kernel of the brute-force algorithm with variation of atomic types in the supercell and energy evaluation (the inner loop in Algorithm 1)—is implemented in C++, while the rest is written in Python. Namely, the Python code part includes such things as varying the supercells and the number of atoms in structures, construction of convex hulls using the pymatgen package<sup>24</sup> and managing input and output data.

At the stage of supercell generation, we do exclude equivalent supercells as described below. The supercell  $S_1$  is equivalent to the supercell  $S_2$  if (1) the vectors of  $S_1$  are linear combinations of the vectors of  $S_2$ , or (2) the vectors of  $S_1$  can be obtained from the vectors of  $S_2$  by orthogonal transformation (see example in Fig. 8). The first condition can be rewritten in the

following form:

$$S_2 S_1^{-1} \in \mathbb{Z}^{3 \times 3}, \quad |\det(S_2 S_1^{-1})| = 1 \quad (5)$$

In our method we generate the set of all non-equivalent supercells for each fixed  $N \in [1, N_{\text{max}}]$  by varying integer numbers of the matrix  $Z$  in (4), that are limited by the first condition of equivalence and the number of atoms in a supercell  $N$ . To verify the second condition of supercell equivalence, we apply all 48 possible orthogonal transformations to each generated candidate supercell and check if it does not coincide with already found ones. If both conditions are satisfied, the candidate is included in the pool of non-equivalent supercells.

As a result of supercell generation, we obtained a set of non-equivalent supercells for each number of atoms  $N$ . The number of such supercells for each  $N$  in the case of BCC and FCC lattices is the same and is provided in Table 2.

It could be seen that the number of non-equivalent supercells grows relatively fast with the number of atoms  $N$  in a supercell. This fact, combined with the number of atomic type combinations for each supercell growing exponentially with  $N$ , imposes restrictions on the performance of our brute force method. Then, for each generated structure using this algorithm, formation energy is evaluated without the need for relaxation (equilibration or energy minimization), since the approach for energy evaluation is based on a machine learning model of interatomic interaction that is preliminary trained to reproduce energies of relaxed (equilibrated) structures. We describe this model in greater detail below.

#### Low-rank potential

The machine learning interatomic interaction model that is used in our research belongs to the class of the so-called on-lattice models, which means that when trained, it can make predictions only for atomistic systems with a single lattice type. It is worth mentioning that, as the system evolves towards its equilibrium state, atoms deviate from their corresponding lattice sites, resulting in minor deformations of the whole lattice. However, when considering on-lattice models, each atom still needs to be associated with some lattice site.

The model of interatomic interaction considered below is based on the low-rank potential (LRP), first published in<sup>25</sup>. LRP belongs to the class of local potentials, which means that the total energy  $E$  of a configuration  $\text{cfg}$  is divided into the sum of the energy contributions of each lattice site  $\xi$  (site

energies  $V$ ):

$$E(\text{cfg}) = \sum_{\xi} V(\xi), \quad (6)$$

where the site energy  $V$  depends only on the type of atom in the site  $\xi$  and the types of atoms assigned to neighboring sites within the cutoff sphere of radius  $R_{\text{cut}}$ . The order of arguments in  $V$  is predefined by the enumeration order of the neighboring sites  $\xi + \mathbf{r}_1, \dots, \xi + \mathbf{r}_m$ , which is unique for all environments of that lattice:

$$V(\xi) = V(\sigma(\xi), \sigma(\xi + \mathbf{r}_1), \dots, \sigma(\xi + \mathbf{r}_m)), \quad (7)$$

where  $\sigma(\xi + \mathbf{r}_n)$  is the atom type in the site with coordinate  $\xi + \mathbf{r}_n$  (the first argument in (7) is the type of the central atom).

The on-lattice interatomic potential  $V$  is fully defined by site energy values, given all possible combinations of atomic types in the central site and neighboring sites. Thus, for an atomistic system with  $m$  components and  $n$  atoms within the cutoff sphere, the on-lattice interatomic potential  $V$  is described by  $m^n$  parameters without taking symmetries into account. Since the potential can also be represented as  $n$ -dimensional tensor (here, we use this term as a synonym for a multidimensional array), it can be approximated by a lower-dimensional mathematical object, which is the core idea of LRP. One example of such low-rank approximations is Tensor Train decomposition (or TT-decomposition)<sup>26</sup>, that may be regarded as an analog of SVD decomposition for multidimensional tensors ( $d > 2$ ):

$$V_{\sigma_1 \dots \sigma_d} = \underbrace{A_1[\sigma_1]}_{1 \times r_2} \underbrace{A_2[\sigma_2]}_{r_2 \times r_3} \dots \underbrace{A_d[\sigma_d]}_{r_d \times 1} \quad (8)$$

where each matrix  $A_i[\sigma_i]$  has dimensions  $r_i \times r_{i+1}$ ; boundary conditions are satisfied:  $r_1 = r_{d+1} = 1$ ; and the rank of such representation is the number  $r := \max r_i$ . This decomposition (8) results in a reduction of the number of model parameters (elements of decomposition matrices) from  $m^n$  to  $mnr^2$  (without taking symmetries into account). The efficiency of TT-representation can be demonstrated by the example of a structure with 4 atomic types and a BCC lattice with 13 neighbors in the first coordination sphere, where the number of parameters is reduced from 67 million to a few thousand. Moreover, changing the maximal rank of TT-decomposition allows tuning the model to find a balance between performance and approximation accuracy.

To find the unknown parameters of the TT-decomposition (i.e., to fit the potential), we minimize the deviation between energies calculated with LRP and those calculated from first principles for each configuration in the training set. Namely, the following functional is minimized:

$$J = \frac{1}{KN_{G_{\text{sym}}}} \sum_{g \in G_{\text{sym}}} \sum_{k=1}^K |E(g(\text{cfg}^{(k)})) - E^{qm}(\text{cfg}^{(k)})|^2 \rightarrow \min \quad (9)$$

where  $E(\text{cfg}^{(k)})$  is the energy of configuration  $\text{cfg}^{(k)}$  predicted by low-rank potential, and  $E^{qm}(\text{cfg}^{(k)})$  is the energy of the same configuration derived from first principles (the total number of configurations in the training set is  $K$ ). Here, we take into account the symmetries of the lattice by applying all linear space transformations  $g \in G_{\text{sym}}$  that leave the lattice unchanged, thus reducing the required training set size.

The minimization problem (9) is solved using the Alternating Least Squares method<sup>27</sup>. First, the initial values of all matrices are specified. Then, the optimal values of each matrix are selected: the elements of matrix  $A_i$  are varied while the values of other matrices  $A_j$  ( $i \neq j$ ) are fixed. Finally, at the end of each iteration, random Gaussian noise is added to the parameter values (the annealing method) to avoid getting stuck in a local minimum. The procedure is repeated until the difference between the values of successive iterations is less than a specified threshold.

## Data availability

POSCAR files containing the geometry of all newly discovered structures, along with the constructed convex hulls for both BCC and FCC systems, can be found and downloaded at: <https://gitlab.com/zinkovich/alloydiscovery>.

## Code availability

The code that was used in the findings of this study is available from the corresponding author upon reasonable request.

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## Author contributions

V.Z. and E.P. trained the ML models, wrote the code, and analyzed the experimental results; V.S. contributed to the code implementation; A.S. and E.P. supervised the work; A.S. contributed to the design of the research; all authors discussed the results and wrote the paper.

## Competing interests

The authors declare no competing interests.

## Additional information

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